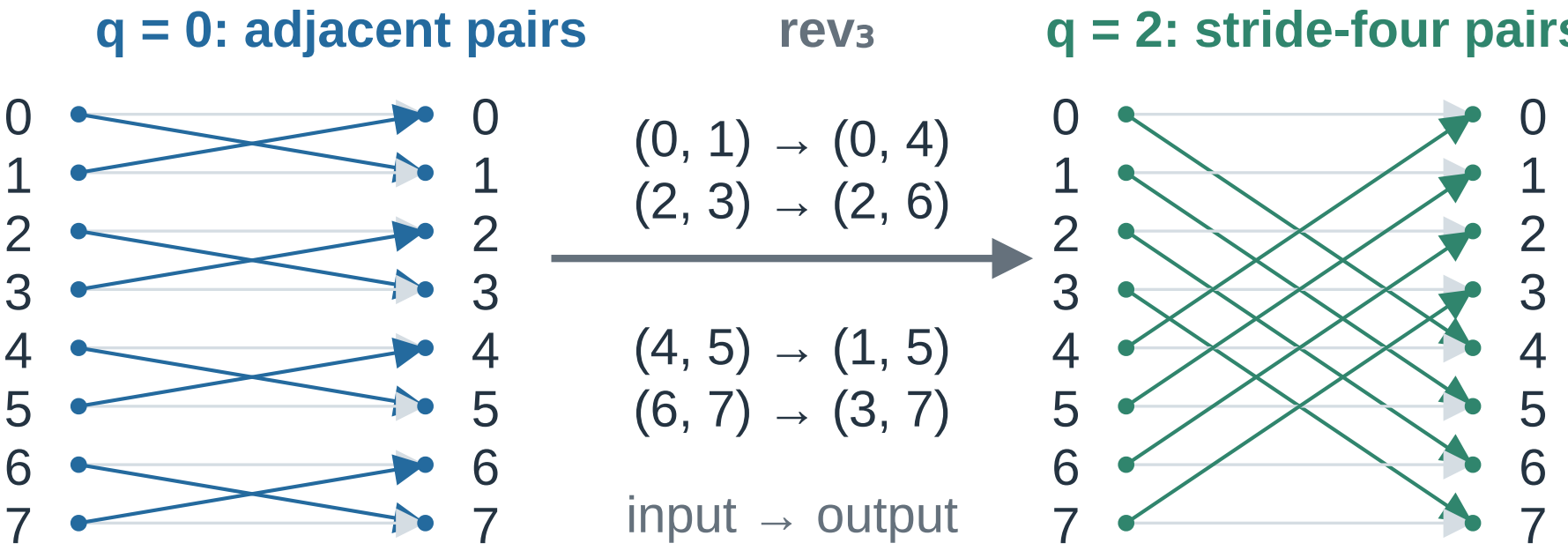
Bit reversal preserves the pairing relation



Carry the ordered pair, coefficient w, and direction:

CT: $(a, b) \rightarrow (a + wb, a - wb)$, before and after rev_3

Whole sequence: $(0, 1, 2) \rightarrow (2, 1, 0)$; labels travel with stages.